

Rishit Sharma

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EDUCATION

Purdue University

Expected May 2028

Bachelor of Science – Computer Science Major – Minors in Quantum Science & Math – 3.84 GPA

- Relevant Coursework: Data Structures & Algorithms (CS 251), Computer Architecture (CS 250)

EXPERIENCE

Co-Founder & Lead Developer

Sep. 2025 – Present

UPlate

- **1,100+ downloads** for a cross-platform Flutter dining application, integrating real-time dietary filtering, generative AI meal recommendations, and simple macro data aggregation to streamline campus dining navigation.
- **Reduced costs by 88%** by architecting a backend migration from Firebase to a hybrid stack (Cloudflare Workers, D1, KV), and allowing future scalability to other colleges.

AI Engineering Researcher

Aug. 2026 – Present

The Data Mine Corporate Partners - Johnson and Johnson

- Architecting an internal data analytics platform combining disparate schema structures across SQL and Graph databases (GraphDB).
- Implementing a Retrieval-Augmented Generation (RAG) pipeline to enable natural language querying and contextual data extraction across enterprise datasets.
- Engineering an interactive analytics dashboard in React/TypeScript to visualize multi-source querying outputs and graph relationships.

PROJECTS & ORGANIZATIONS

Computer Vision Developer

Sep. 2025 – Present

Autonomous Motorsports @ Purdue

- **Cut curve-fit error by 36.6%** by building the perception pipeline for a go kart in collegiate autonomous racing, integrating DBSCAN clustering into an OpenCV segmentation pass before RANSAC curve fitting (**4.49** → **2.85** px RMS).
- Expanded the autonomous sensor suite by integrating LiDAR for all-condition obstacle detection and deploying an Android-based mobile sensor device as a redundant IMU/GPS failover subsystem, boosting localization reliability.

Technical Mentor

Aug. 2026 – Present

Launchpad Purdue

- Mentoring first-year CS students through end-to-end full-stack development lifecycles, code architecture reviews, and git version control systems.

Portfolio Website | React, Framer Motion, GitHub

Summer 2025

- Developed a React website to showcase other projects, utilizing Framer Motion for fluid animations.
- Utilized GitHub as a database for authorized writes and unlimited fast reads.

WeatherML | Python, PyTorch, NOAA Data

Summer 2024

- Used **10 years** of historical weather data from the NOAA to train a feedforward neural network with PyTorch.
- $\pm 3^{\circ}\text{F}$ accuracy after training the model to predict the weather in Clive, Iowa.

HONORS AND CERTIFICATIONS

AWS Certified Developer - Associate (DVA-C02)

July 2026

Purdue University Dean's List Recipient

Fall 2025 & Spring 2026

TECHNICAL SKILLS

Programming Languages: Python, Dart, C, Java, JavaScript/TypeScript, R, SQL, HTML/CSS

Frameworks/Runtimes: Flutter, React, Tailwind, workerd, PyTorch

Libraries: OpenCV, Pandas, NumPy, Matplotlib, Framer Motion

Cloud: AWS (Lambda, S3, DynamoDB, API Gateway, Eventbridge), Cloudflare Workers/D1/KV, Firebase, REST APIs

Conversational Languages: English (Native), German (Conversational), Hindi (Conversational)

Tools: git/GitHub, Claude Code, Codex, Zed, GitHub Actions, Postman, Jupyter Notebooks